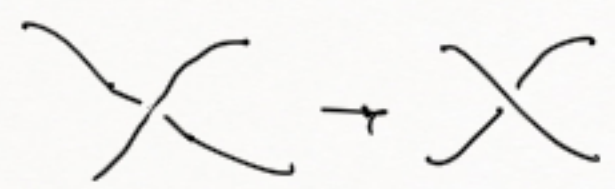


Finding invariants

Unknotting number



Least number of unknottings to trivialize a knot
 $\Rightarrow$ Knot inv. indr.

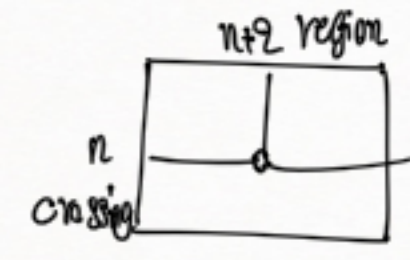
1923 Alexander



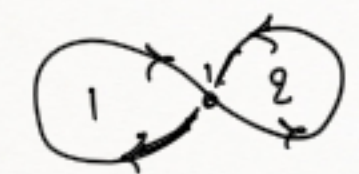
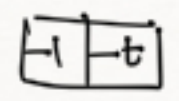
σ_i, σ_i^{-1}
Any knot is the closure of a braid!
Braid index

1928 Alexander poly.

n crossing
 $\Rightarrow$ n+2 regions



not adj = 0
region on left before undercross $\Rightarrow -t$
" " " after " $\Rightarrow t$
" " right before " $\Rightarrow 1$
" " " after " $\Rightarrow -1$



Remove 2 cols corresponding to adj. region

Det unique up to mult by $\pm t^n \Rightarrow$ Normalization: Constant term > 0

$\Rightarrow$ --- Jones

Mazur: Cov transform acts on X , $H_1(X) \Rightarrow H_1(X)$ is $\mathbb{Z}[t, t^{-1}]$ -mod.

$\Rightarrow$ Finitely representable, torsion $r \leq g$ gens rel

$8 \times r$ matrix $\Rightarrow$ Ideal gen. by $r \times r$ minors $\Rightarrow$ Alexander ideal

$\Rightarrow$ Principal, gen. unique up to mult by $\pm t^n$

$\Rightarrow$ This is the same as det of $H_1(X)$ as $\mathbb{Z}[t, t^{-1}]$ -mod.

p prime, $\mathbb{Z}_{p^n}$ is p^n -th PRU $\Rightarrow D_n := \mathbb{Z}[\mathbb{Z}_{p^n}]$, $\mathbb{Z} \hookrightarrow \mathbb{Z}[\mathbb{Z}_{p^n}]$

$\Rightarrow \text{Spec } \mathbb{Z}[\mathbb{Z}_{p^n}] \hookrightarrow \text{Spec } \mathbb{Z}$ cyclic Galois cover ram. only at prime p

Fraction field $K_n = \mathbb{Q}[\mathbb{Z}_{p^n}] \Rightarrow$ inf tower $K_0 \subseteq K_1 \subseteq K_2 \subseteq \dots$

$K_\infty = \cup K_i$; cyclotomic $\mathbb{Z}_p$ -ext of $K_0 = \mathbb{Q}[\mathbb{Z}_p]$

Galois group $\Gamma = \text{Gal}(K_\infty/K_0)$

X_n p -primary part of ideal class group of D_n , Norm maps $N_{n+1}: X_{n+1} \rightarrow X_n$ inv. system.

$\Rightarrow X := \varprojlim X_n$

$\Rightarrow X$ has the structure $\hat{\mathbb{Z}}_p[[T]]$ -module

By Iwasawa th.
 $\Rightarrow$ Torsion module of finite type

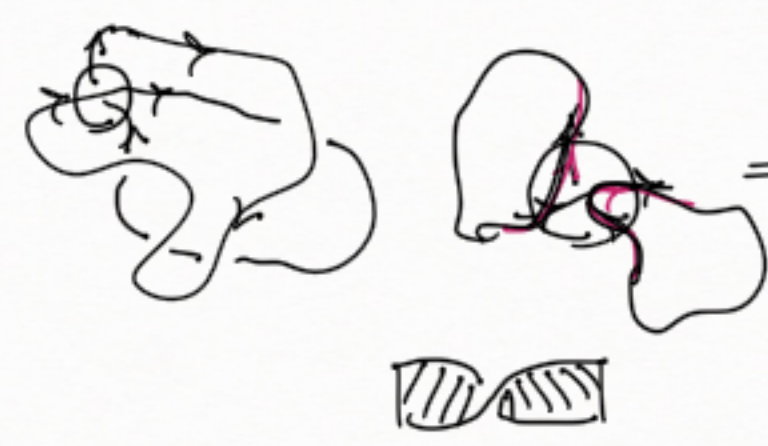
Addition: Connected sum



$\Rightarrow$ Well-defined

Prime knot: Knot K s.t. $K = K_1 \# K_2 \Rightarrow K_1$ or K_2 unknot

Seifert surface: Surface in S^3 com, comp, knot as boundary



- i) Eliminate crossings
- ii) Left with a disjoint union of closed curves
- iii) Fill with disks
- iv) Fix with twists
- v) Connect with tubes

Genus of surface
Genus of knot = min genus of a Seifert surface

$g(K_1 \# K_2) = g(K_1) + g(K_2)$

K knot $\Leftrightarrow g(K) = 1$

Schubert: $K_1 \# K_2 = K_2 \# K_1$

$(p_1, \dots, p_n) + 1$ $\Rightarrow$ not finite

Whitehead double $\Rightarrow$ Genus of WH double of nontrivial knot $\Rightarrow$

$K_1 \text{ prime} \text{---} K_n \Rightarrow (K_1 \# \dots \# K_n)$ WH double $\Rightarrow$ New prime!

$\mathbb{Z}$
mult comm.
1
FTD
prime, FTA
=

$qa = qb \Rightarrow a = b$
Euclid Lemma
 $plab \Leftrightarrow ab = pk$
 $\Rightarrow p|a \Leftrightarrow a = pk' \Rightarrow k'b = k$
or $p|b \Leftrightarrow b = pk' \Rightarrow k'a = k$

Knots
comm. sum. comm.
unknot
 X
prime, Schubert
Amb iso./Ori. pres. homeo.
 p prime, $p \# K_1 = p \# K_2 \Rightarrow K_1 = K_2$
Euclid? Lemma
 p prime, $p \# K = K_1 \# K_2$
 $\Rightarrow K_1 = p \# K', K' \# K_2 = K$
or $K_2 = p \# K', K' \# K_1 = K$

Unknotting number $\Rightarrow$ Open until 2025

$\tau_1 \neq -\tau_1$
Crossing number $\Rightarrow$ Open

$\mathbb{Z}/n\mathbb{Z}$ Prime ideals $\mathbb{Z}/p\mathbb{Z}$
$$V(I) = \{P \mid I \subseteq P\}$$
$$V(\mathbb{Z}/n\mathbb{Z}) = \{p \in \mathbb{P} \mid p \mid n\}$$

$\Rightarrow$ Co-finite topology

Étale topology/cohomology

Presheaf over $\mathcal{C}$: Contravariant functor $\mathcal{C} \rightarrow \underline{\text{Sets}}$
 Ab

Sheaf: i) $\forall U \subseteq X$ open, $\text{cov } \{U_i\}_{i \in I}$, $\forall s_i \in \mathcal{F}(U_i) \Rightarrow s|_{U_i} \equiv t|_{U_i} \Rightarrow s \equiv t$
 ii) $\forall U \subseteq X$ open, $\text{cov } \{U_i\}_{i \in I}$, $\forall s_i \in \mathcal{F}(U_i) \Rightarrow s_i|_{U_i \cap U_j} \equiv s_j|_{U_i \cap U_j} \Rightarrow \exists s \in \mathcal{F}(U) : s|_{U_i} \equiv s_i$

Site: Cat $\mathcal{C}$, $\forall U \in \text{Obj } \mathcal{C}$ maps $\{U_i \rightarrow U\}_{i \in I}$ cov of U

i) $\forall (U_i \rightarrow U)$, mor. $V \rightarrow U$ in $\mathcal{C} \Rightarrow U_i \times_U V$ exists $U_i \times_U V \rightarrow V$ covering of V

$$ii) \bigvee (V_i \rightarrow U) \text{ cov. of } U, \{V_{ij} \rightarrow U\}_{i \in I_i} \text{ cov. of } U_i \Rightarrow (V_{ij} \rightarrow U)_{i \in I_{\cup I_i}} \text{ cov. of } U$$

ii) $\forall U \in \text{Obj } \mathcal{C} \Rightarrow U \xrightarrow{\text{id}} U$ is a cov.

Étale site over X : Objects $\mathcal{E}t/X$ Étale morphisms $U \rightarrow X$, $\forall U: (U_i \rightarrow U) \text{ Étale}$

$$X_{\text{ét}} \Rightarrow \text{Sh}(X_{\text{ét}}) \text{ Ab. cat. with enough inj.}$$
$$\mathbb{F} \mapsto \Gamma(X, \mathbb{F}) \text{ left exact } \text{Sh}(X_{\text{ét}}) \rightarrow \text{Ab.} \Rightarrow n\text{-th right derived functor } H_{\text{ét}}^r(X_{\text{ét}}, -)$$

Sheaf $\mathcal{F} \Rightarrow$ Inj. res $\circ \rightarrow \mathcal{F} \rightarrow \mathcal{I}_0 \rightarrow \mathcal{I}_1 \rightarrow \dots$

$$\Gamma(X, \mathbb{F}) \rightarrow \Gamma(X, \mathbb{F}_2) \rightarrow \Gamma(X, \mathbb{F}_1) \rightarrow \dots \quad \text{not exact}$$
$$\left(\begin{array}{l} H^0(X_{et}, F) = \Gamma(X, \chi_{et}) \\ F \text{ inj} \Rightarrow H^r(X_{et}, F) = 0, \forall r > 0 \\ 0 \rightarrow F' \rightarrow F \rightarrow F'' \rightarrow 0 \\ \rightarrow H^0(X_{et}, F') \rightarrow \dots \end{array} \right.$$

Étale dim: Least N st $H^n(X_{\text{ét}}, \mathbb{F}) = 0, \forall n > N, \forall \mathbb{F}$ torsion sheaf

Ann-Verdier duality: $H^r(X, F) \otimes \text{Ext}^{3-r}(F, G_m) \rightarrow H^3(X, G_m) = \mathbb{Q}/\mathbb{Z}$

CFT considerations $\Rightarrow$ Compactify $\text{Spec } \mathbb{Z} = X$

$$((\hat{X}_{\text{ét}}, \text{cov. } (F, F_0)) \xleftrightarrow[\substack{\text{Étale sheaf} \\ \text{over } X = \text{Spec } \mathbb{Z}}]{=} \text{Gal}(G_p)\text{-mod structure} \\ \text{comp. with } F$$

over $X = \text{Spec } \mathbb{Z}$

Coh. theory satisfy $\dots \rightarrow \underbrace{H^i(\hat{X}_{\text{ét}}, F)}_{\text{Étale cohomology}} \rightarrow H^i(X_{\text{ét}}, F) \rightarrow \hat{H}^i(\text{Gal}(\bar{\mathbb{C}}/\mathbb{C}), F_{\infty}) \rightarrow H^{i+1}(\hat{X}_{\text{ét}}, F) \rightarrow \dots$

$$\frac{\dim_{\mathbb{C}t}(\hat{X}_{\mathbb{C}t}) = 3}{\sqrt{\quad}} \Rightarrow H_{\mathbb{C}t}^4(\hat{X}_{\mathbb{C}t}, \mathbb{F}) = 0$$
$$H^3_{\text{ét}}(\hat{X}_{\text{ét}}, \mu_n) = \mathbb{Z}/n\mathbb{Z} \neq 0$$
$$0 \rightarrow \mu_n \rightarrow G_m \xrightarrow{\text{power of } n} B_m \rightarrow 0$$
$$\begin{aligned} \cdots \rightarrow H^2_{\text{ét}}(\hat{X}_{\text{ét}}, G_m) \rightarrow H^3_{\text{ét}}(\hat{X}_{\text{ét}}, \mu_n) \rightarrow H^3_{\text{ét}}(\hat{X}_{\text{ét}}, G_m) \rightarrow H^3_{\text{ét}}(\hat{X}_{\text{ét}}, G_m) \rightarrow \cdots \\ \underbrace{\hspace{10em}}_{\cong \mathbb{Z}/n\mathbb{Z}} \quad \underbrace{\hspace{10em}}_{\substack{\mathbb{Q}/\mathbb{Z} \\ \text{mult by } n \\ \mathbb{Z}/n\mathbb{Z}}} \quad \underbrace{\hspace{10em}}_{\mathbb{Q}/\mathbb{Z}} \end{aligned}$$
$$\begin{array}{c} 8^1 \hookrightarrow 8^3 \\ \text{Knot} \\ \hline \end{array}$$

$\pi_1(\widehat{\text{Spec } \mathbb{Z}}) = 0$ $\swarrow$ Simply connected

$$\pi_1^{\text{ét}}(\text{Spec } \mathbb{Z}/n\mathbb{Z}) = \left(\hat{\mathbb{Z}} \right) \leftarrow \text{Analogue of } \mathbb{Z} \Rightarrow \text{Spec } \mathbb{Z}/n\mathbb{Z}$$

Poincaré duality, Simply connected $\Rightarrow$ Homology groups of 3-manifold agree with homologies of S^3

$\Rightarrow$ Consider map $f: S^3 \rightarrow X$ of deg 1 $\Rightarrow$ A map of simp. conn. CW-comp inducing iso. of all homologies exists since $H_3(X) = H_3(S^3) \cong \mathbb{Z}$ is a homotopy equiv. $\Rightarrow$ Homotopic to S^3

Poincare conj. in 3-dim $\Rightarrow$ Homeomorphic to S^3